Experiment 2: Insertion Sort – Card Sorting While Playing

Aim

To implement the **Insertion Sort** algorithm in Python to sort playing cards as they are picked one by one.

Problem Statement

A card player picks up cards one at a time and inserts each new card into its correct position in the hand. Implement this behavior using **Insertion Sort**.

Algorithm

Step 1: Start.

Step 2: Read the number of cards.

Step 3: Read the card values.

Step 4: Apply Insertion Sort by inserting each card into its correct position.

Step 5: Display the sorted hand.

Step 6: Stop.

Source Code (Python)

main.py

```python
# Card Sorting While Playing using Insertion Sort

n = int(input("Enter number of cards: "))
cards = []

print("Enter card values:")
for i in range(n):
    cards.append(int(input()))

for i in range(1, n):
    key = cards[i]
    j = i - 1

    while j >= 0 and cards[j] > key:
        cards[j + 1] = cards[j]
        j -= 1

    cards[j + 1] = key

print("Sorted Cards:", cards)
```

Output

```
Output

Enter number of cards: 5
Enter card values:
7
2
9
4
1
Sorted Cards: [1, 2, 4, 7, 9]

=== Code Execution Successful ===
```

Complexity Analysis

Aspect	Complexity
Time Complexity	$O(n^2)$
Space Complexity	O(1)

Result

The Insertion Sort algorithm was successfully implemented to sort the cards. Each card was inserted into its correct position, producing the sorted hand correctly.